

EE/CE 252/251

Signals and Systems

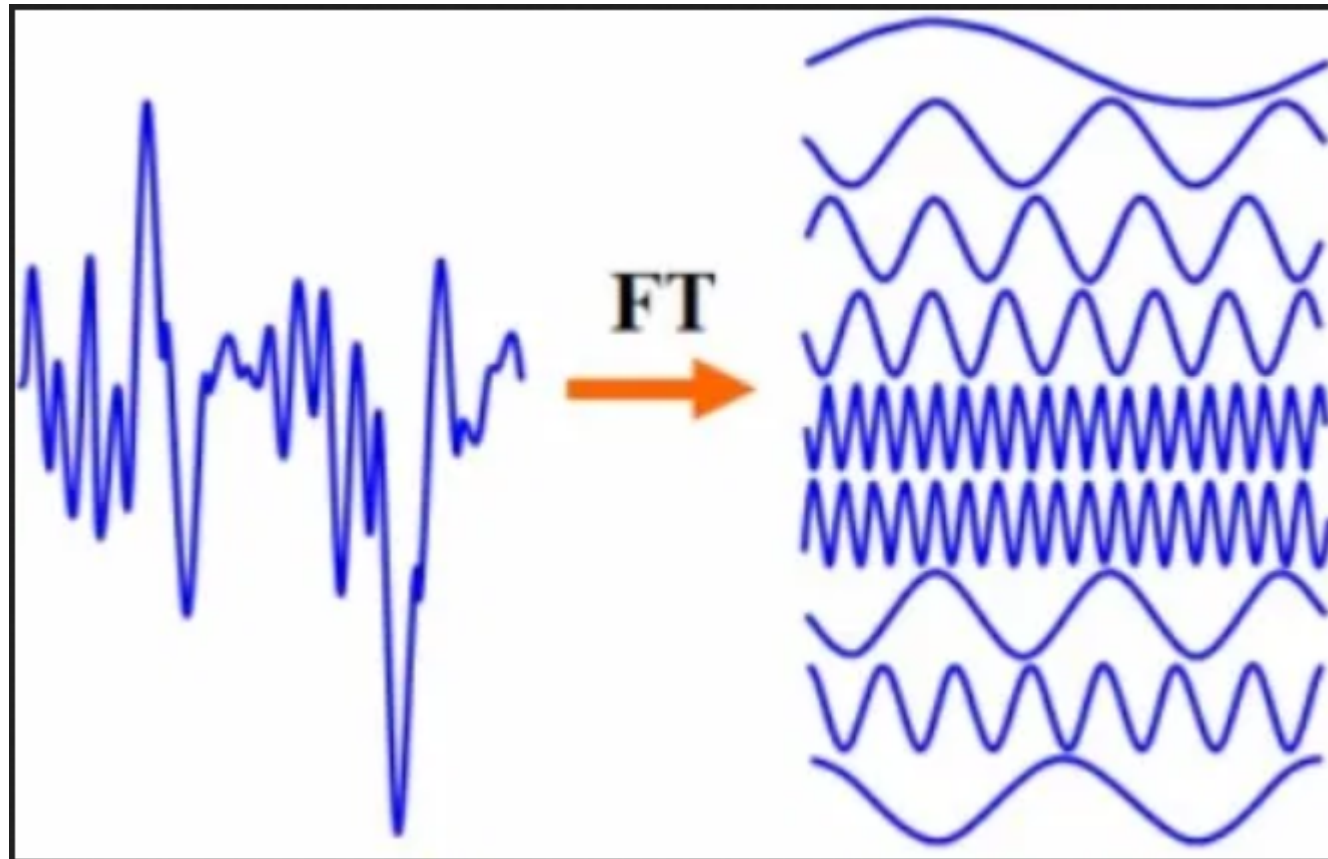
with

Dr. Naveed R. Butt

@

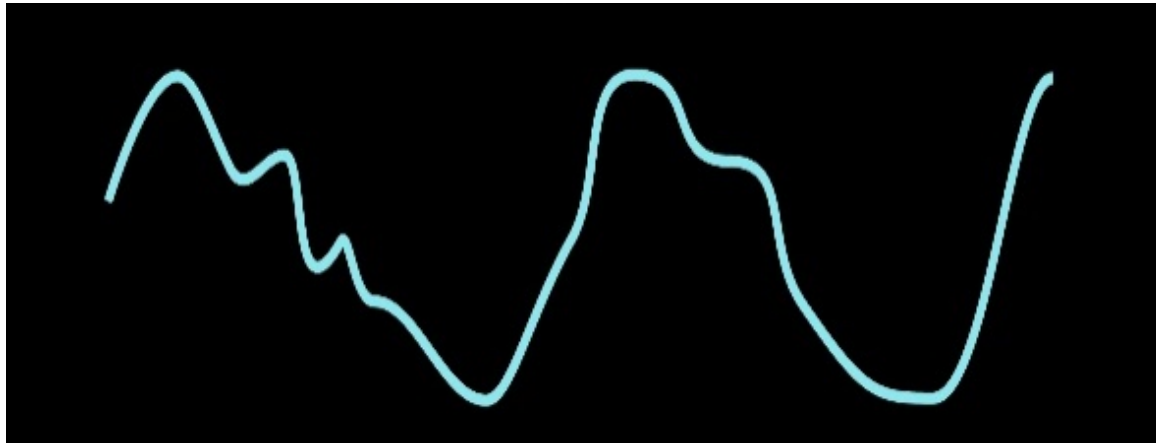
Habib University

Q. Can we write signals as sums of periodic functions (frequencies)?

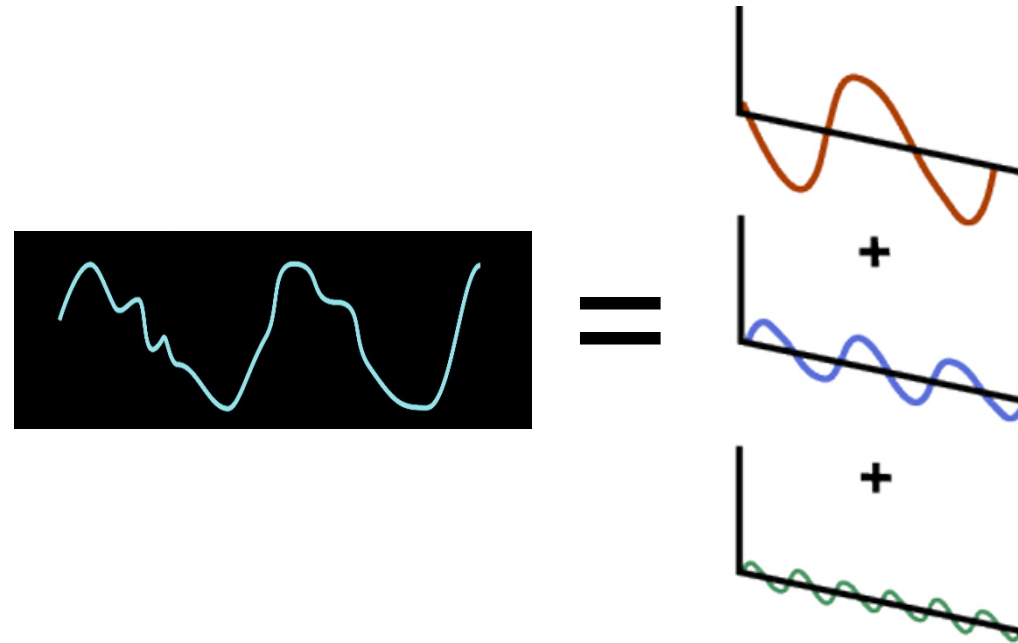


Baking a Fourier Cake

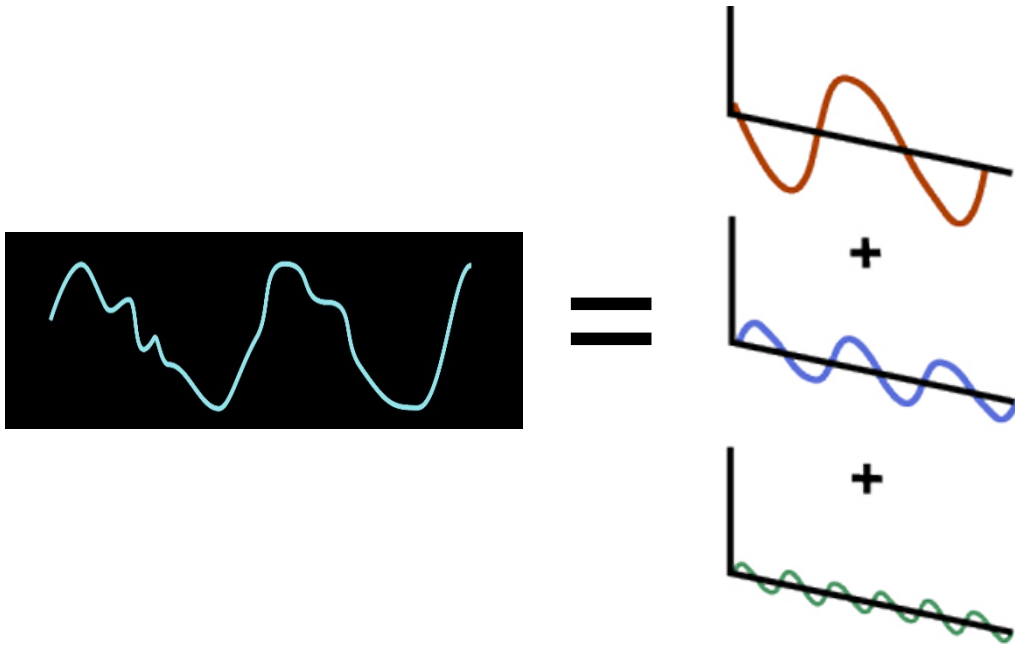
- **Given:** Signal shape (time-domain)
- **Ingredients:** Sinusoids of different frequencies
- **Choose:** How much of the each ingredient (sinusoid) to use?



- In Fourier Transform, we want to look at signals in terms of a fixed set of ingredients
 - Ingredients : Sinusoids of different frequencies

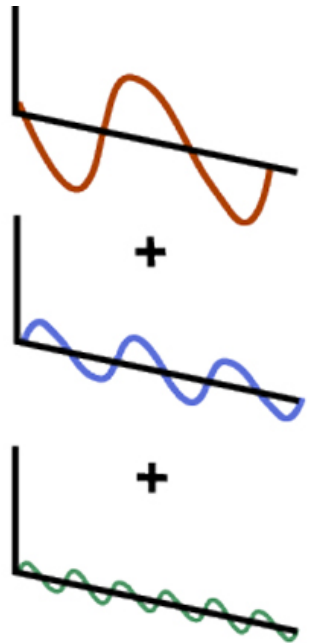


- In Fourier Transform, we want to look at signals in terms of a fixed set of ingredients
 - Ingredients : Sinusoids of different frequencies

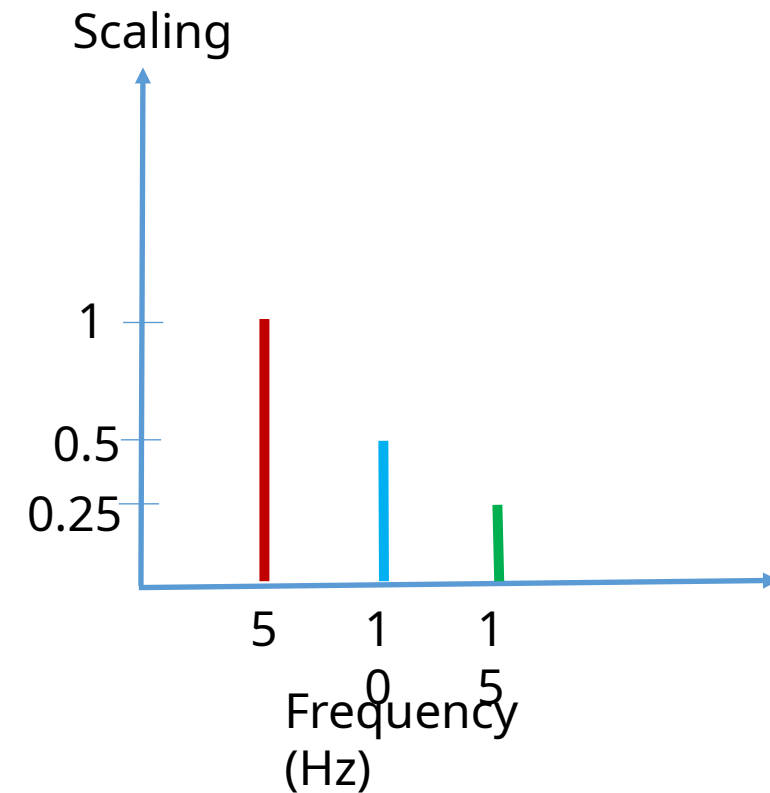


Ingredient (sinusoid frequency)	Amount (scaling)	Process
f_1	1	Add all
f_2	0.5	
f_3	0.25	

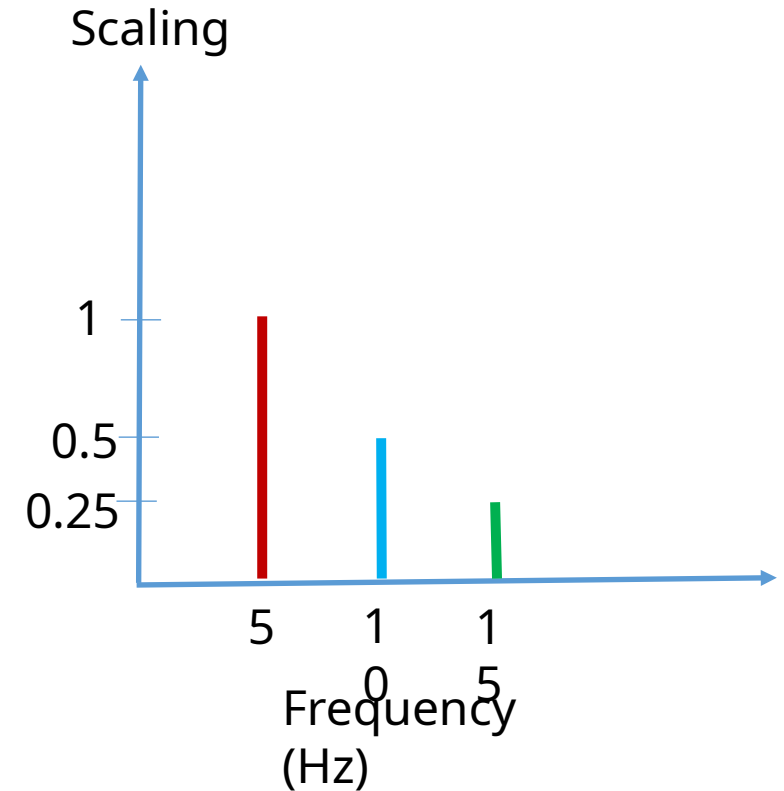
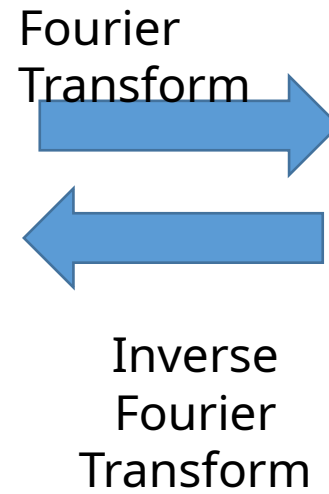
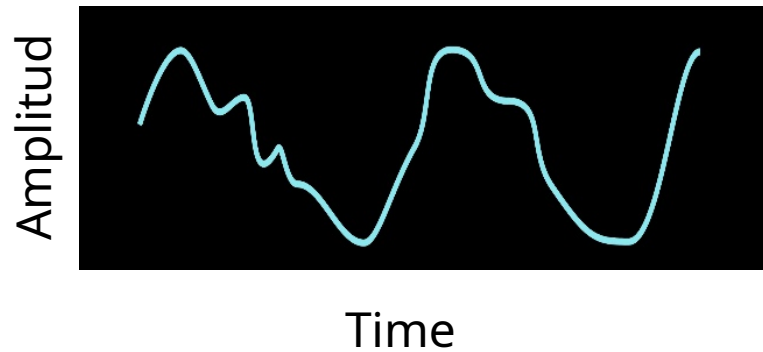
- How is this shown after Fourier transform?



Ingredient (sinusoid frequency)	Amount (scaling)	Process
5 Hz	1	Add all
10 Hz	0.5	
15 Hz	0.25	

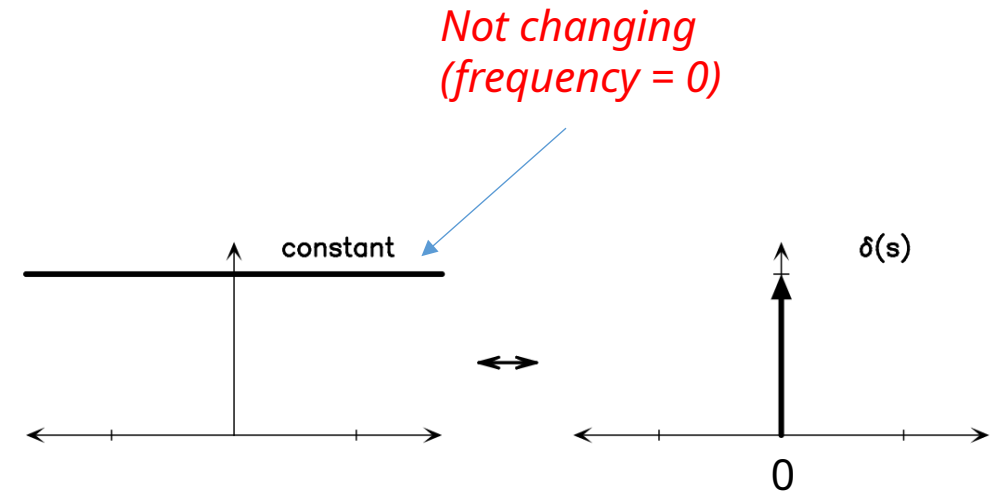
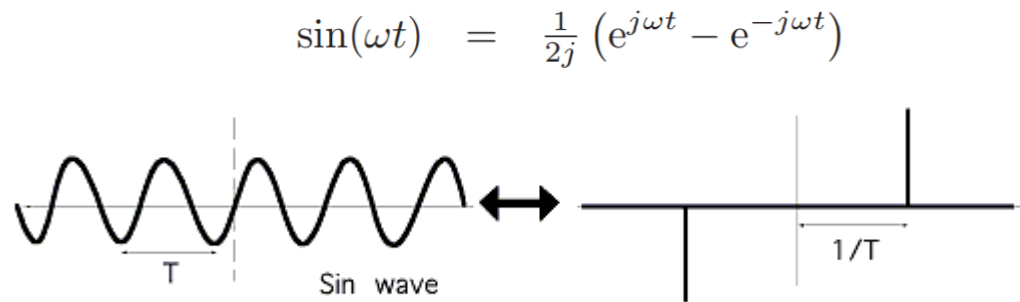


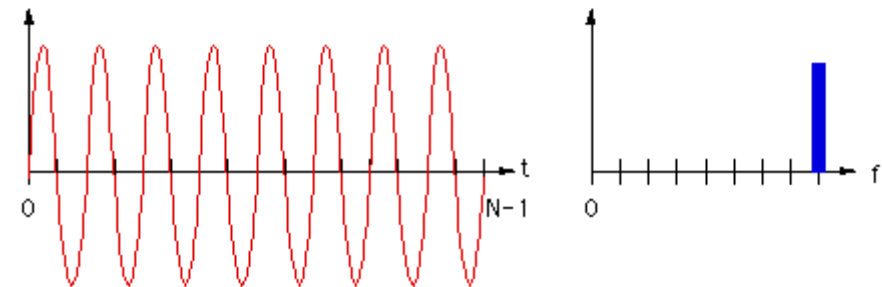
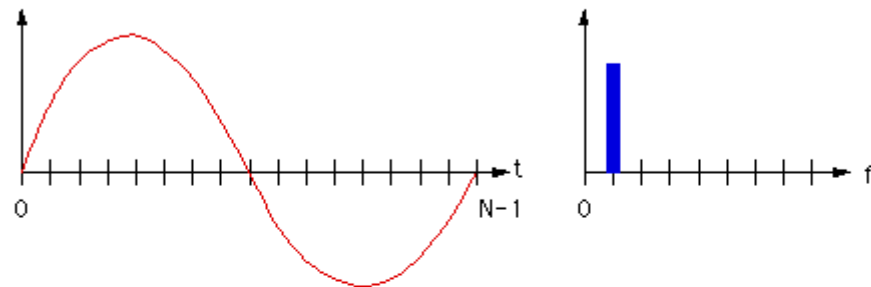
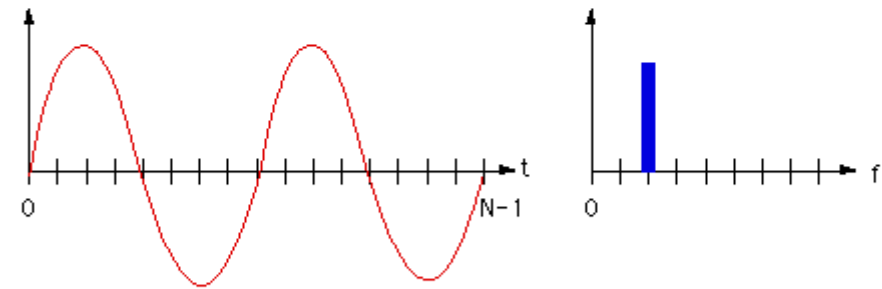
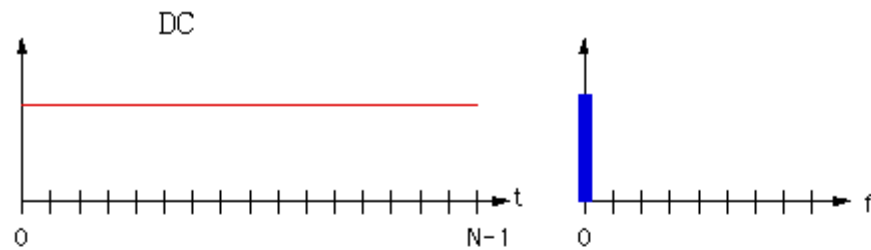
- We mostly skip the middle steps



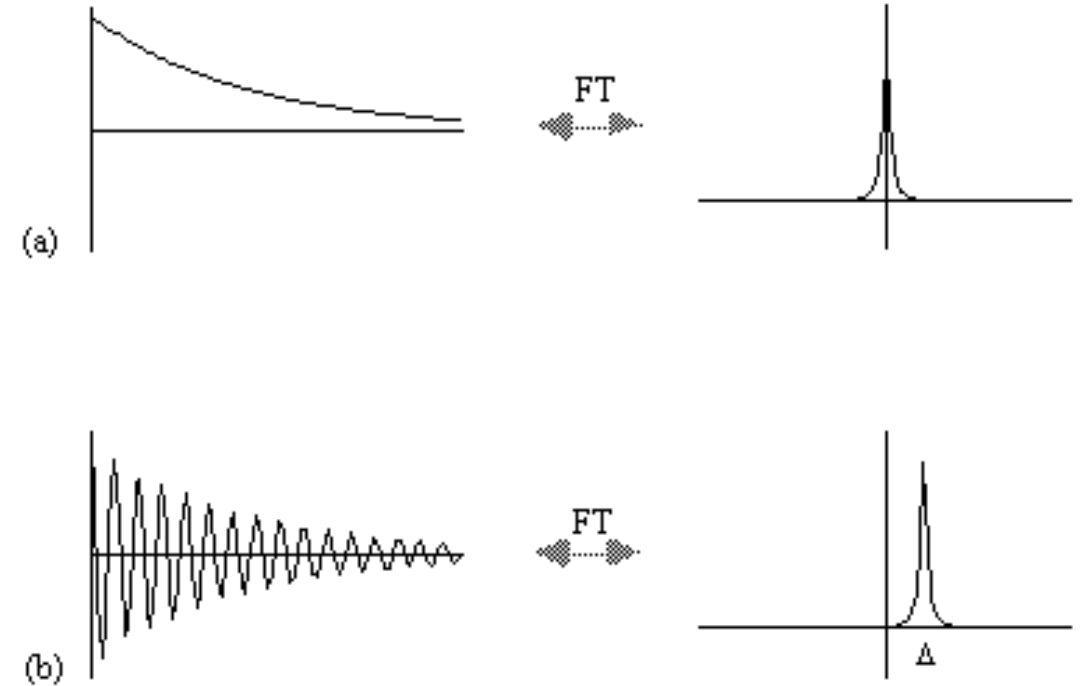
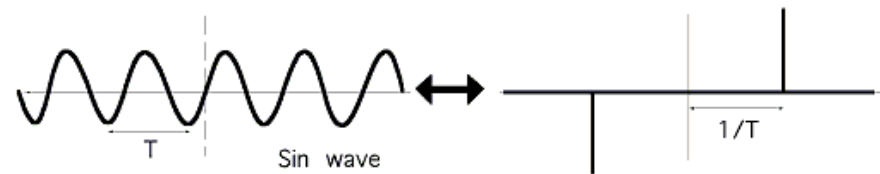
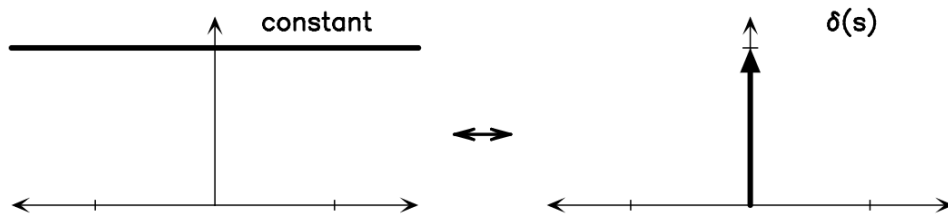
FT – Pairs and Rules-of-Thumb

Some Fourier Transforms (Visual)

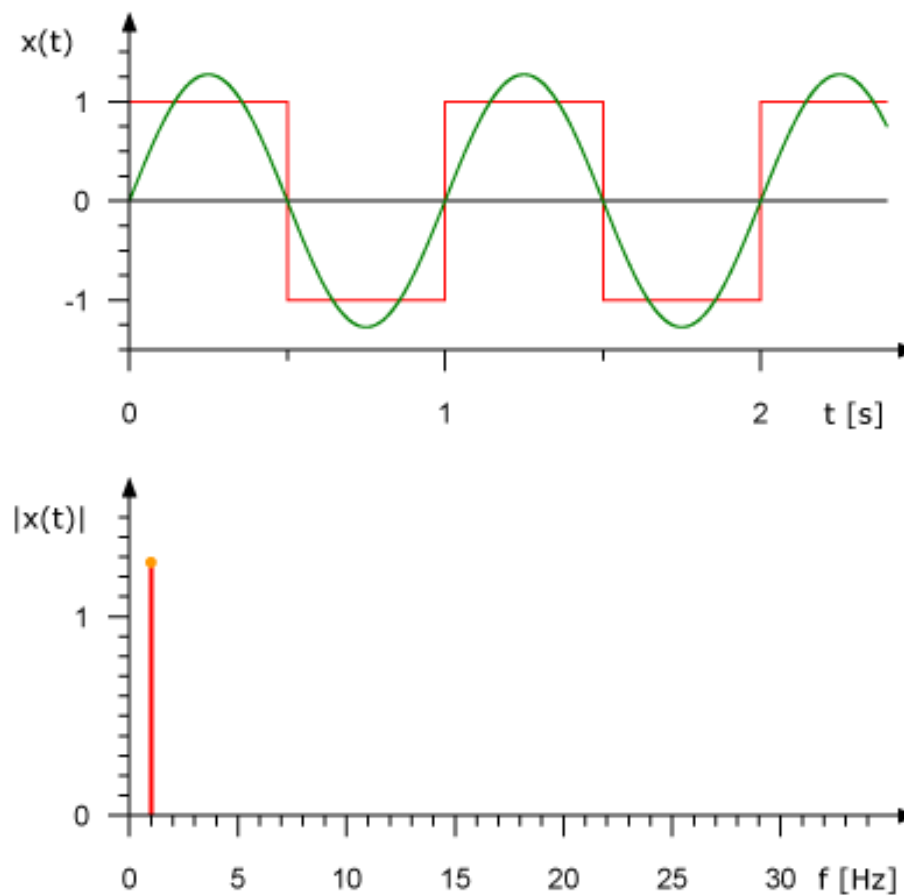
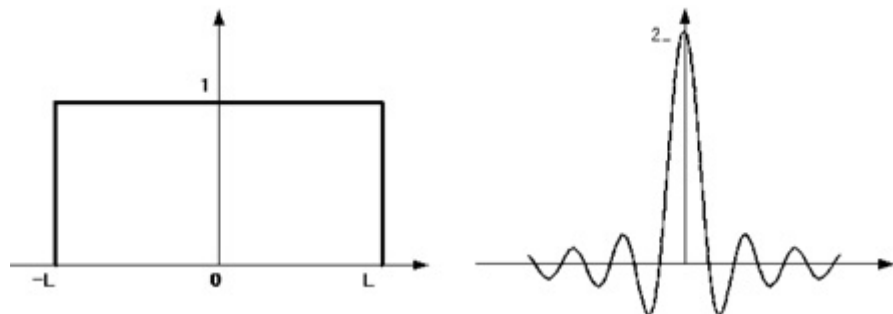




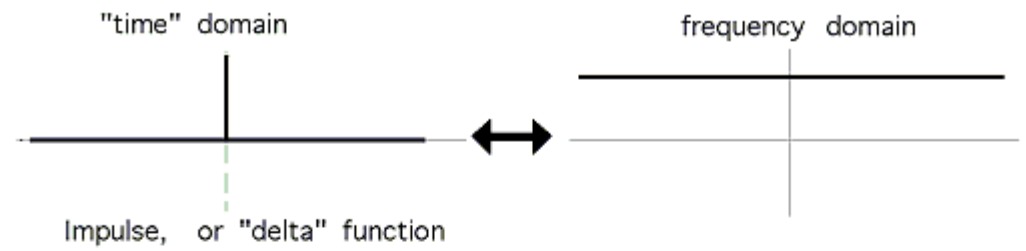
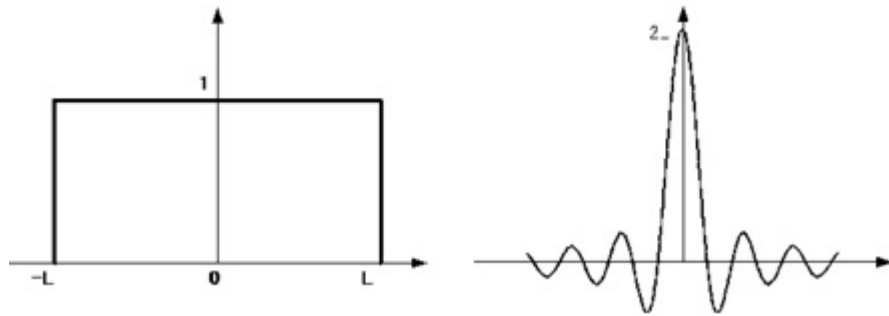
Rule1: As frequency increases, the FT peaks move outwards



Rule 2: Damping causes spread

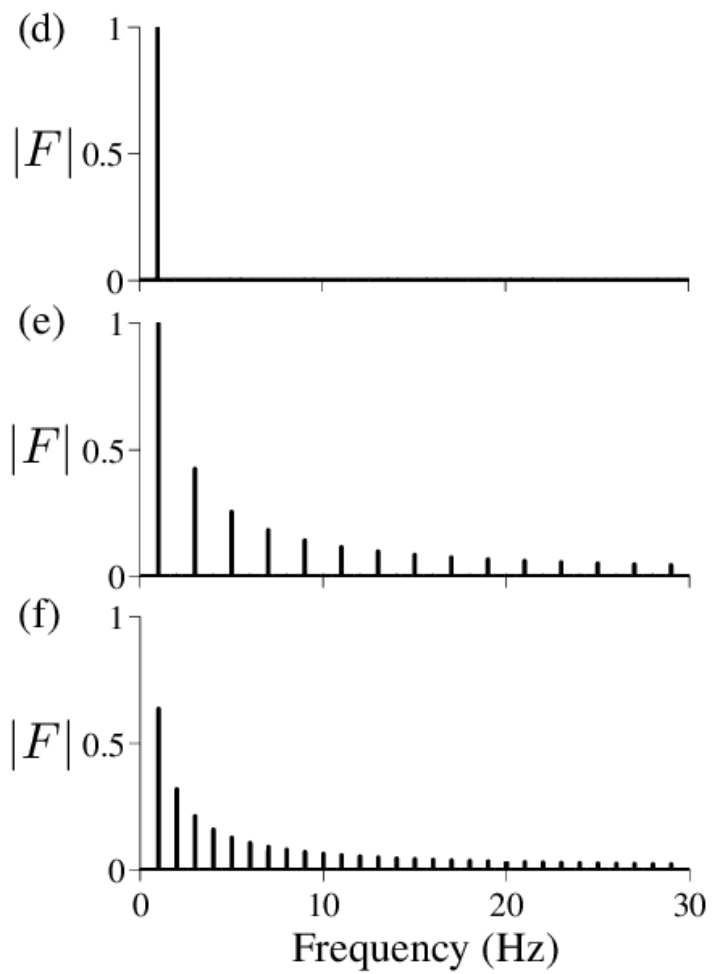
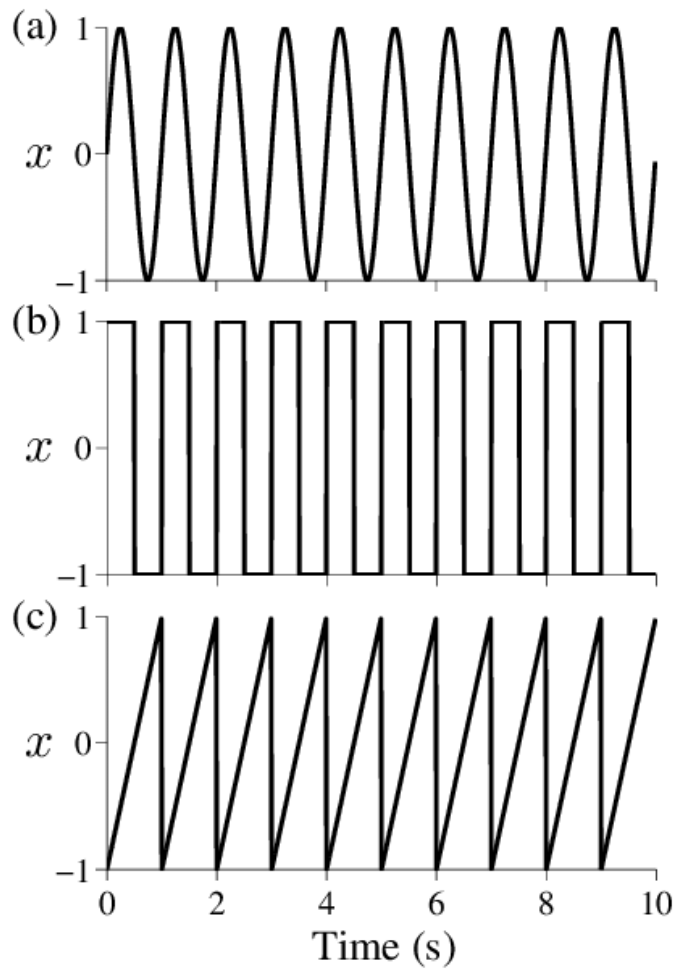


Rule 3: Sharp changes (edges) require a lot of frequencies

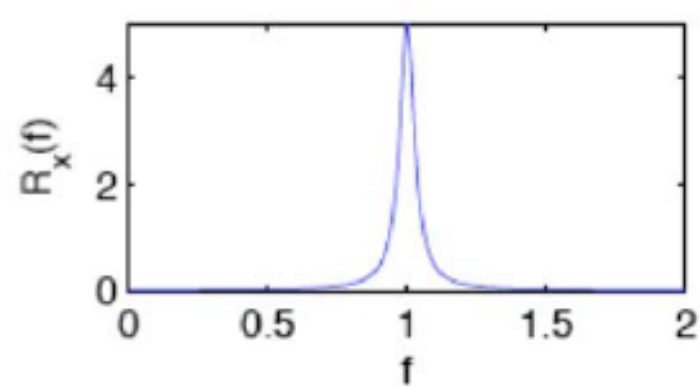
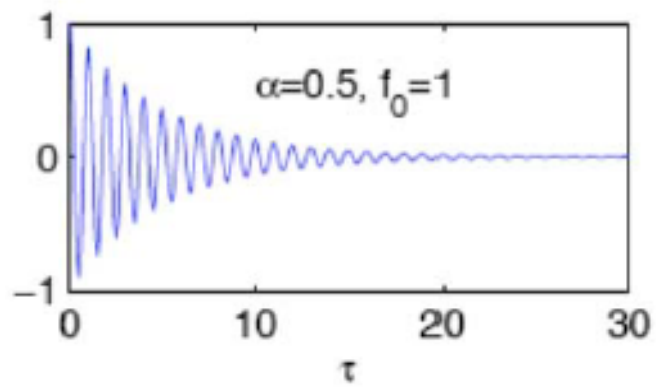
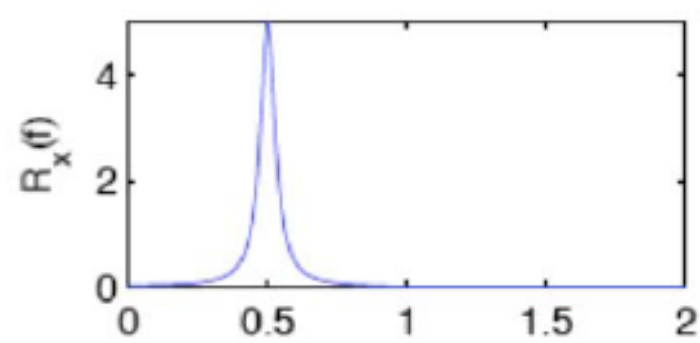
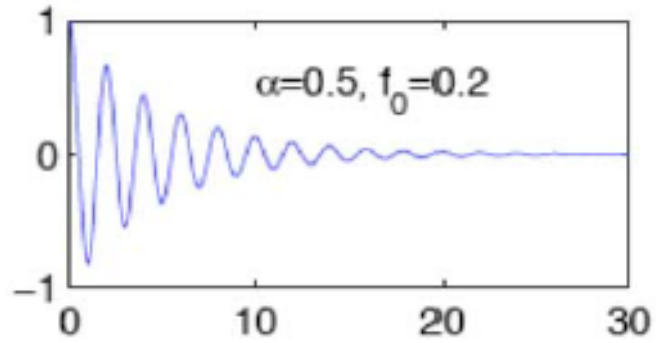
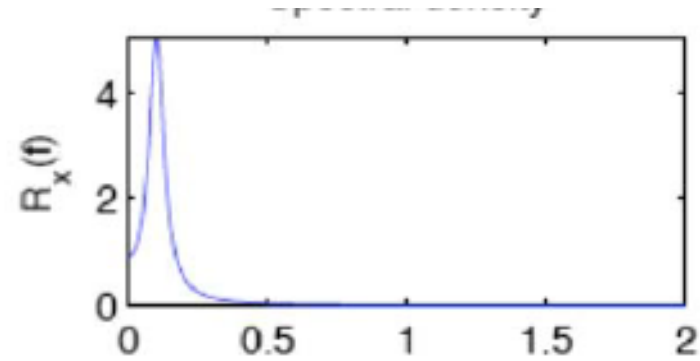
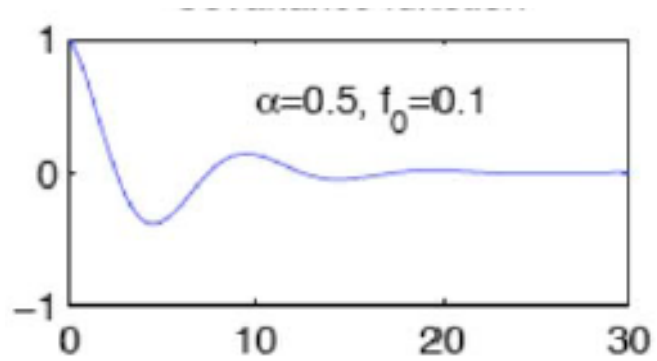


... and an extremely sharp change (impulse) requires ALL the frequencies!!

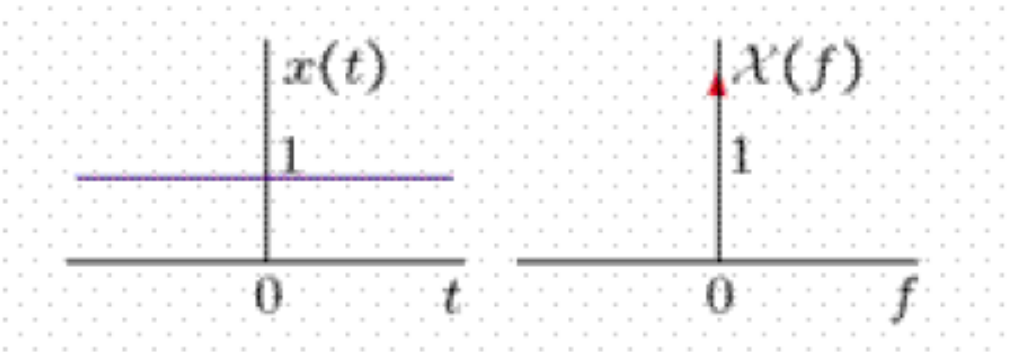
Rule 3: Sharp changes (edges) require a lot of frequencies



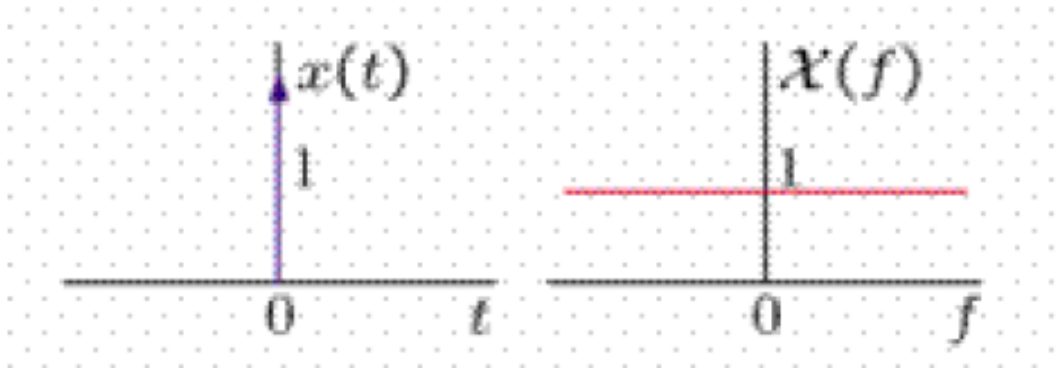
Rule 4: Periodic functions have discrete spectra.



Rule 5: Multiple effects can be combined.



Rule 6: Duality



Questions?? Thoughts??

